14_ popquiz magnetism

This is a preview of the published version of the quiz

Started: Mar 22 at 7:13pm

Quiz Instructions

Question 1 17 pts

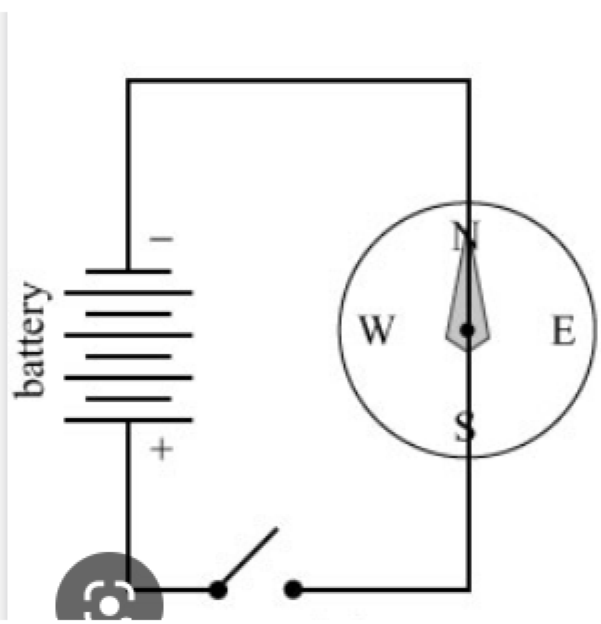
What is true about electric fields/magnetic fields

- both magnetic and electric fields only diverge/converge from/to monopoles
- magnetic fields only form loops , electric fields can diverge/converge from/to monopoles
- both magneltic and electric fields only form loops
- magnetic fields converge/diverge to/from monopoles, electric fields only form loops

Question 2 17 pts

In class, I did a demo with a compass and a DC circuit (9V battery, switch, connecting wires).

The needle of the compass was parallel to one connection wire and the switch was initially open.

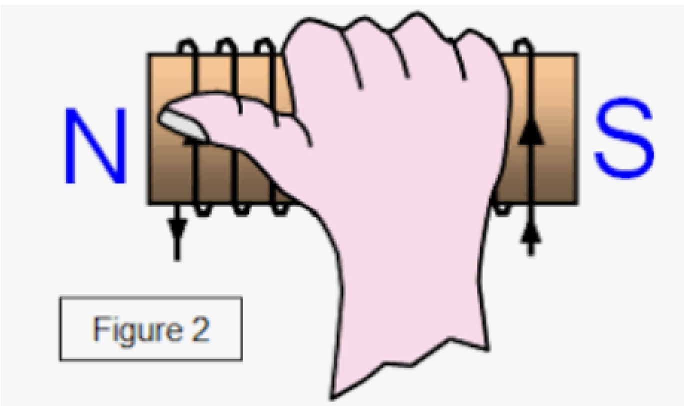


When I closed the switch, what happened to the needle?

- It started to oscillate non stop back and forth
- It didn't move and stayed parallel to the wire
- It moved and positionned itsel perpendicular to the wire
- I missed the class

Question 3 17 pts

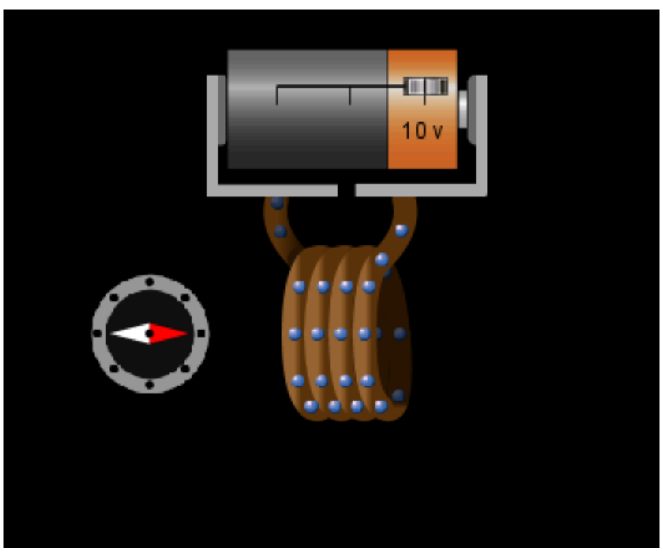
You an use the right hand rule to find which side of an electromagnet is North. The thumb shows you the north



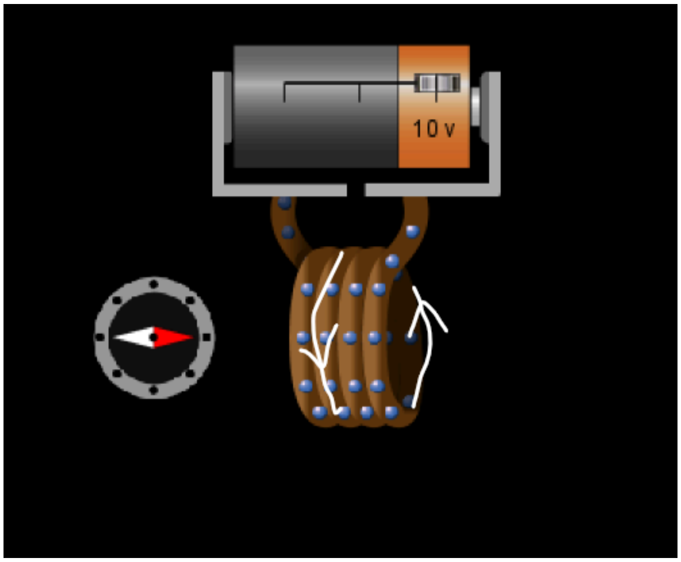
Remember a compass is a magnet itself. The red side shows the south (North attracted to South).

so the red part of a compass is the North.

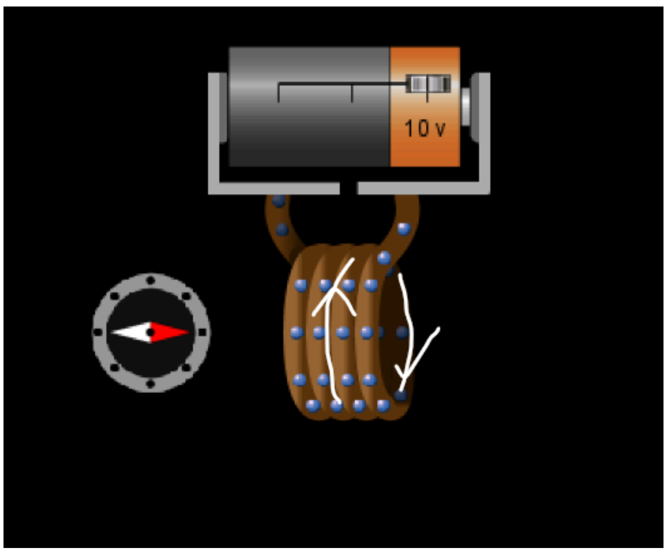
In this figure, Which way the current is flowing



A)



B)



○
A

○
Whats a compass

○
electricity can not produce magnetism

○
B

Question 4 17 pts

AS discussed in class,

magnetic fields loop

○
From North to South

○ don't loop but converge toward a monopole

○ From South to North

○ Don't loop but diverge toward a monopole

Question 5 17 pts

The same way you can find electric monopoles (eletrons, protons . .) , you can find magnetic monopoles (isolated North or South poles)

○ True

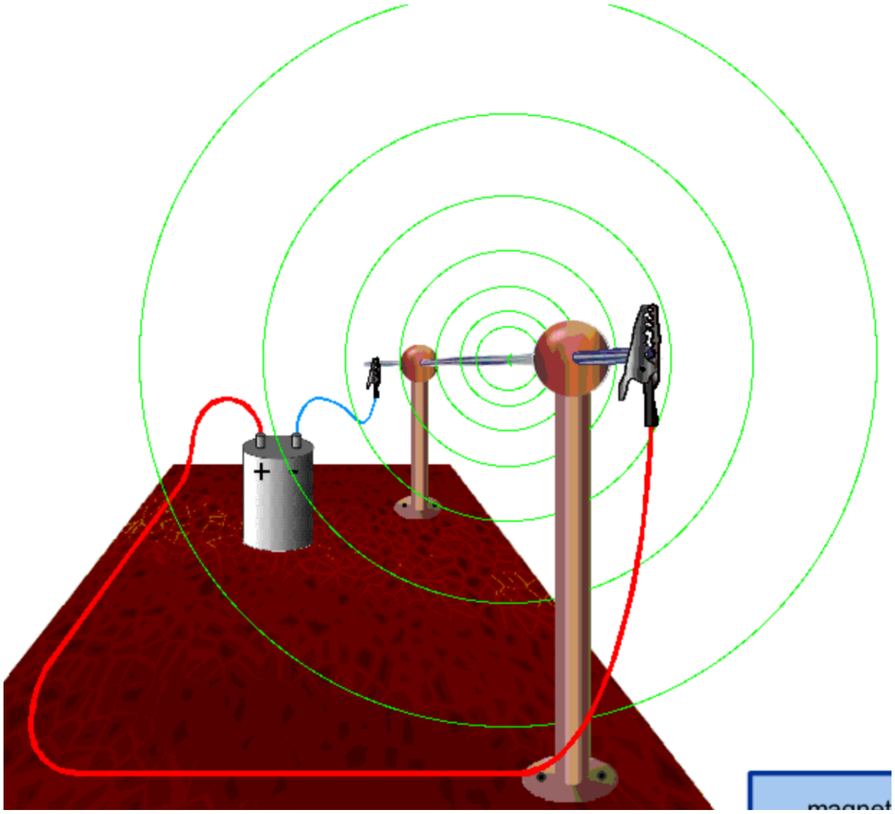
○ You can not find electric monopoles and you can not find magnetic monopoles

○ I missed the class

○ False

Question 6 17 pts

You can use the right hand rule for another purpose. Finding which way the magnetic field loop around an electric wire as shown below:



In the figure above you can see the terminals of the battery so you know the direction of the current in the solid wire (going in the screen). The mgnetic field looping around the wire is going

- ◯ CW
- ◯ electricity does not produce magnetism
- ◯ I am going to miss the bus
- ◯ CCW

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